rsna-knee-abnormality-detection

3rd Place Solution: RSNA Knee Abnormality Detection 2026

Rank	3
Team	DataTech Club
Author	Zhihong Li
Topic ID	752510
Version	Original

Source: https://www.kaggle.com/competitions/rsna-knee-abnormality-detection/discussion/752510
Retrieved with Kaggle CLI on 2026-10-27.

We thank **RSNA and Kaggle**, the participating institutions and radiologists, and the open-source community for making this competition possible.

This solution is a joint effort by **Zhengru Li, Zhihong Li, Guoqin Gu, Mutian Hong, and Bufan Li**. We are members of **DataTech**, a student club at **ShanghaiTech University** dedicated to data science competitions, machine learning, and AI exploration. We warmly welcome fellow competitors, researchers, and potential collaborators to connect with us. Feel free to learn more about our club at **DataTech Club**!

9th on the public leaderboard (0.968) · 3rd on the private leaderboard (0.959)

Overview

Only **58 of 4,407 training studies** had expert-adjudicated labels. The remaining studies had multilingual reports, but the evaluation target was an image-based diagnosis. We built **multimodal soft labels**, trained **2.5D MRI readers**, and combined complementary configurations with **group-aware probability blending**. Reports and annotation models were used only to construct supervision; deployed readers used MRI alone.

Submission	Design	Public	Private
G36-Gold	**36 checkpoints; grouped probability blend**	**0.968**	**0.959**
A21GT	26 checkpoints; three-expert hard routing	0.968	0.958

G36-Gold produced our third-place result. The two selected submissions were evaluated separately, not averaged. All leaderboard scores below use Kaggle's three-decimal display.

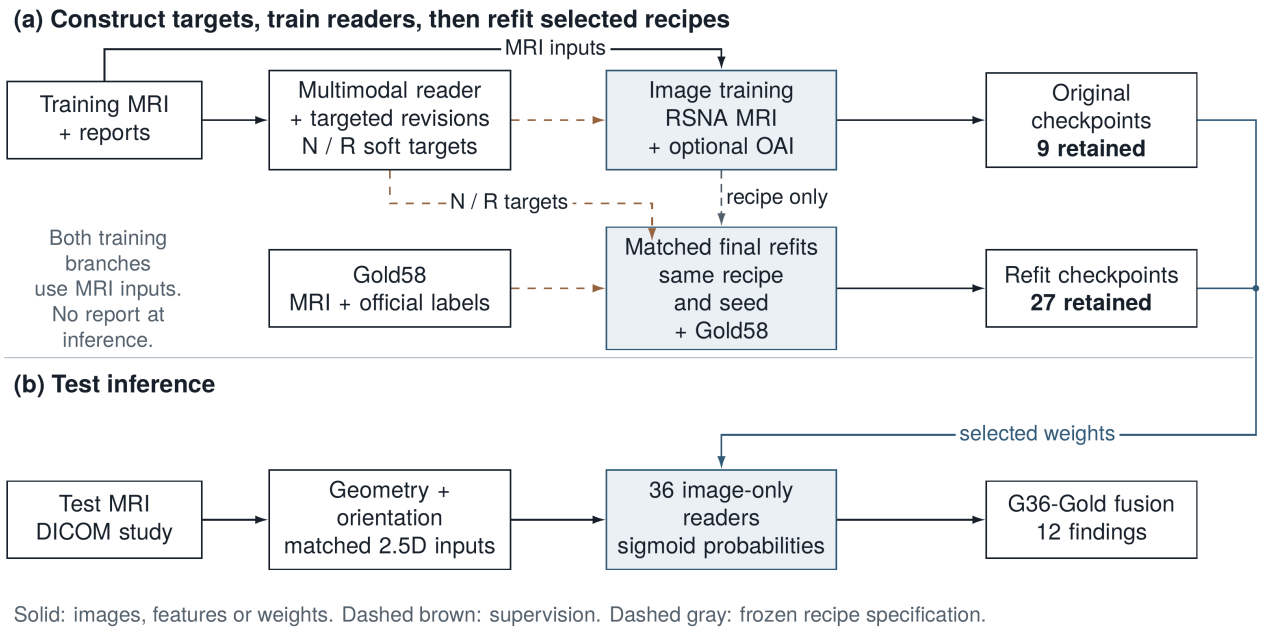


Figure 1. Training and inference. Frozen recipes lead to matched Gold58 refits or retained original checkpoints; the dashed gray connection transfers recipes, not learned weights.

1. Building supervision from imperfect reports

1.1 Read the images, not just the diagnostic terms

Reports can omit findings or describe changes below the competition's positive threshold. We asked **GPT-6 Astra to review MRI and the report** and assign 12 continuous targets. Finding-specific readers then refined **PF OA, Lateral OA, and Synovitis**.

For the PF specialist, ordered sheets held up to nine sampled 384-px slices from one plane/sequence group. The reader examined images first, checked location, severity and extent, then reconciled the report. Other report-reader opinions were supporting evidence, not votes. Accepted revisions changed only the relevant targets. Students trained on frozen snapshots, without reports or reader explanations.

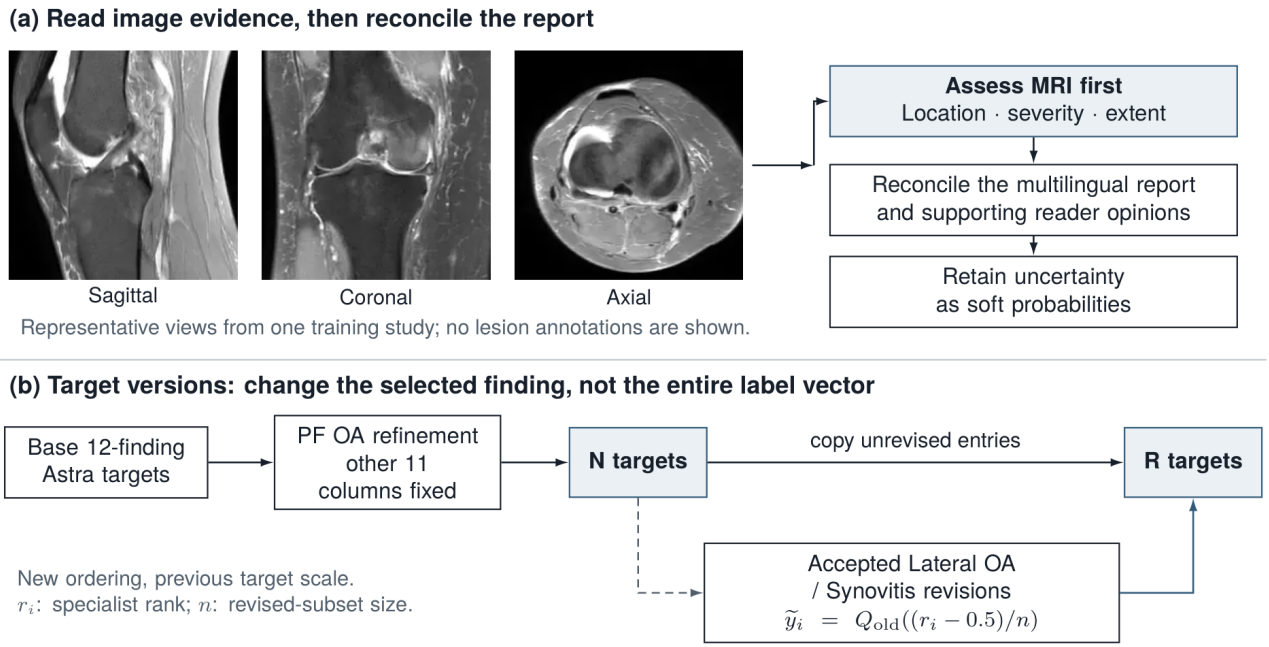


Figure 2. From image evidence to target versions. The three real MRI views come from one training study's frozen input cache; they illustrate the image format, not a saved annotation session or a diagnostic finding. The lower panel shows the actual target-construction rule.

1.2 Preserve complementary targets

N is the PF OA-refined target set. **R** additionally uses specialist rankings for accepted Lateral OA / Synovitis revisions. Within each revised subset, ranks select quantiles of the previous targets, preserving the new ordering while limiting scale changes. Unrevised entries remain fixed. A new reader can change both ordering and probability scale; quantile mapping separates those effects. This is applied only inside each accepted revision subset, not to predictions across the test set. Both versions enter the model pool; **R denotes a target transformation, not a ranking loss**.

1.3 External OAI supervision

Selected readers also trained on MRI from the **Osteoarthritis Initiative (OAI)**, from a pool of approximately **9,440 knees**. Its expert readings do not completely match the competition targets. The main v3kd50 recipe

mixes applicable expert targets and ten-model teacher targets 50:50 on reliably mapped cells; other cells retain their fill values or reduced weights.

Before mixing, the teacher probability is deweighted as `t=p/[p+w_c(1-p)]`, where `w_c` is its positive-class BCE weight. Some runs also apply an OAI-only prevalence correction to training logits, absent at test time. G36-Gold includes **13 RSNA-only readers** and **23 OAI-mixed readers**, with different per-epoch OAI budgets (Section 7).

2. Reading MRI across planes and slices

2.1 Geometry and local context

Slices are sorted by **DICOM physical geometry**, knee orientation is standardized, and missing views are masked. The five acquisition groups are sagittal fluid-sensitive, sagittal non-fluid-sensitive, coronal fluid-sensitive, coronal non-fluid-sensitive, and axial. A roughly **140 mm field of view** is resized to 224, 384 or 512 px. Legacy and native single-resampling pipelines remain paired with their original checkpoints. A representative cache allocates 18, 14, 12, 8 and 12 center positions to these groups respectively, up to 64 in total, with missing positions masked. Each checkpoint retains its own preprocessing and sampling configuration.

A **2.5D window stacks three immediately adjacent source slices from one series**, as shown below. Each window is encoded by BiT ResNetV2-50, ConvNeXt-Small, DINOv3 ViT-B/16 or CoAtNet. Class-specific spatial pooling, global average pooling and attention MIL aggregate the window evidence into 12 study logits.

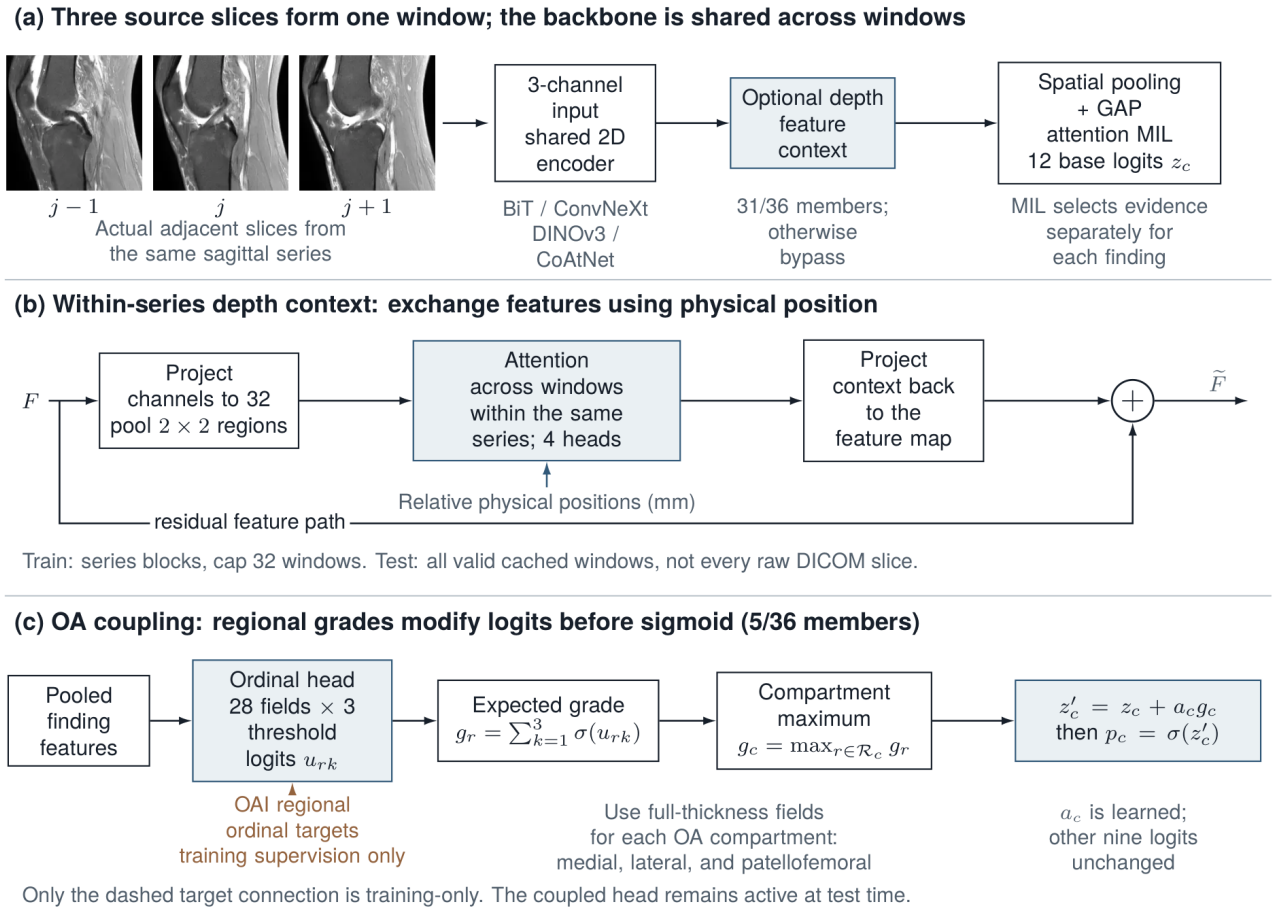


Figure 3. What the reader computes. (a) Actual adjacent sagittal slices form one three-channel input. (b) Optional within-series depth context exchanges features using physical position, with a residual connection. (c) The OA-coupled head converts ordinal outputs to compartment evidence and modifies three logits before sigmoid. Feature paths are schematic, not measured attention maps.

2.2 Depth attention and OA coupling

Depth context (31/36 members) uses 32-dimensional projections, 2×2 regional pooling and four attention heads within each series. Its training bags preserve series blocks, capped at 32 windows. Inference covers all valid cached windows, not every raw DICOM slice.

OA coupling (5/36 members) learns regional cartilage grades, not lesion boxes. Figure 3c shows how cumulative ordinal predictions become compartment-wise maxima and learned residuals for the three OA logits. Other nine logits stay unchanged. OAI grades supervise training only; the coupled prediction head remains active at test time.

3. Training and configuration diversity

We use **positive-weighted soft-target BCE** with mass-preserving normalization: inverse target-dependent positive-weight mass, scaled by its class-wise training mean. Targets stay continuous. Members use a **fixed 25-epoch endpoint**, not the best Gold58 epoch. For example, the RK4C recipe configures OneCycle with a nominal 50-epoch horizon and applies an epoch-18-to-25 tail ending at 2% of peak learning rate; training stops at epoch 25. Individual members retain their recipe-specific schedule settings. Final members are full-data recipe/seed variants, not one uniform five-fold ensemble.

Component	Typical setting
Optimization	AdamW; backbone LR 3e-5 to 1e-4; head LR 1e-3
Schedule	OneCycle-based; epoch-18-to-25 annealing tail
Effective batch / weight decay	Typically 8 / 0.01
Augmentation	Intensity, affine, scale and gamma; recipe-specific interpolation
Checkpoints	EMA 0.999 or fixed-endpoint Raw
Inference	FP16; chunked window encoding

G36-Gold contains **17 BiT, 7 ConvNeXt, 7 DINOv3 and 5 CoAtNet checkpoints**. Resolution counts are **4 / 23 / 9** at 224 / 384 / 512 px; readouts are **32 EMA and 4 Raw**. These counts do not determine the final weights.

Diversity with an explicit budget

Matrix20 allocates five configurations to each backbone family, covering resolution, N/R targets, OAI dose and input processing. It is a curated set, not a full factorial search. Exact entries are in the appendix.

(a) Model memberships by group

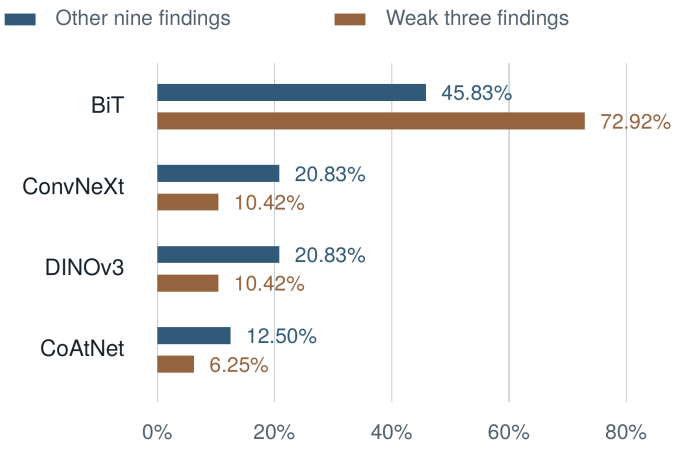
Counts include the Mature12/Expert8 overlap.

	BiT	CNX	DINO	CoAt
Mature12	8	2	2	0
Matrix20	**5**	**5**	**5**	**5**
Expert8	8	0	0	0

Only Matrix20 gives equal budget to each backbone. Mature12 and Expert8 deliberately favor BiT recipes.

Four shared checkpoints are computed once. Their two group contributions remain in the blend.

(b) Effective weight in each output



Linear blend coefficients; not model counts or causal attributions.

Figure 4. Effective weights after expanding the group formula. Matrix20 is backbone-balanced; the full ensemble deliberately favors BiT, particularly on the weak three. These are linear mixing coefficients, not causal feature attributions.

This structure prevents the number of experiments or seeds for one architecture from automatically determining its share. It does **not** make the full ensemble backbone-balanced: Mature12 emphasizes mature BiT recipes, and Expert8 consists entirely of BiT variants. Configuration differences provide candidate diversity, not proof of independent errors.

4. G36-Gold: fixed group budgets, finding-specific support

Three groups serve different roles: **Mature12** anchors the prediction with strong established recipes; **Matrix20** supplies balanced backbone coverage; **Expert8** adds RK4/RK4C-family support for **Lateral OA, PF OA and Synovitis**.

The experts generally remain twelve-output models; “expert” describes which predictions receive extra ensemble weight, not necessarily a single-class training objective. Likewise, Matrix20 balances architecture shares within its branch rather than optimizing a separate weight for every checkpoint.

For sigmoid probabilities `p[m, c]`, the base prediction is:

```
B[c] = 0.5 * mean(p[m, c] for m in Mature12)
     + 0.5 * mean(p[m, c] for m in Matrix20)
```

The other nine findings use `B[c]`. The weak three use:

```
P[c] = 0.5 * B[c] + 0.5 * mean(p[m, c] for m in Expert8)
```

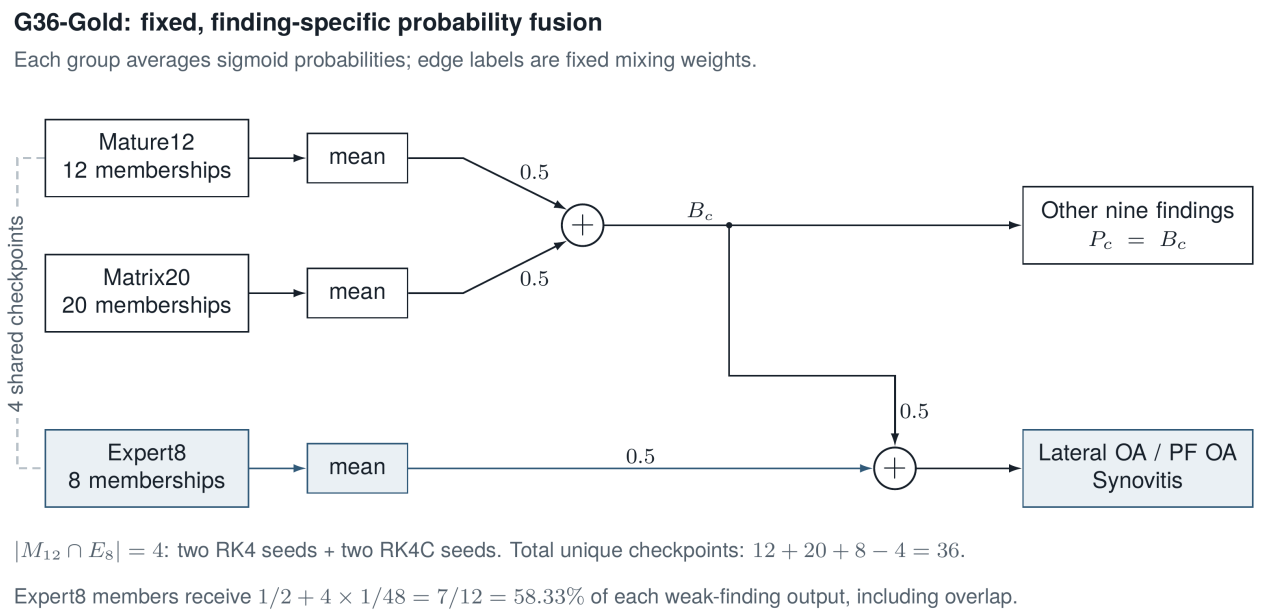


Figure 5. G36 fusion. Four shared checkpoints are inferred once but retain their contributions through both Mature12 and Expert8. Fusion uses probabilities, without percentile-rank conversion.

Two RK4 seeds and two RK4C seeds belong to both Mature12 and Expert8, giving **12 + 20 + 8 − 4 = 36 unique models**. The eight expert members collectively receive **58.33%** of expanded weak-finding weight, including their Mature12 contributions.

Final Gold58 refits

After freezing G36, we replaced eligible members with refits including the **58 expert-labeled studies**, matching recipe, seed, input processing and readout. Unmatched members retained their original weights: **27 refits + 9 original checkpoints**, with group membership and fusion fixed.

Gold58 had already informed prompt and model development. We did not use post-refit, in-sample Gold58 scores to reselect members or weights; those predictions served execution/parity checks. Section 7 separates development scores from retrospective leaderboard results.

5. Alternative submission: hard-routed A21GT

A21GT keeps **21 A21 members** for an equal probability mean on nine findings. Each weak finding instead receives an independent three-expert mean, with no base-blend contribution:

Finding	Routed specialists
Lateral OA	U5 tissue-aware RK4, RK4C, ConvNeXt-512 R
PF OA	ConvNeXt-512 R + OAI, RK4C-lib, ConvNeXt-512 R + full OAI
Synovitis	RK4Call, RK4C (seed 2027), RK4 (seed 2027)

For the recipe names used here, **RK4** is the BiT-50 depth reader with R targets and 4,000 OAI knees per epoch; **RK4C** adds the OA-coupled head. **RK4C-lib** uses an alternative OAI PF-OA target/weight mapping, **RK4Call** uses the full OAI pool per epoch, and **U5** adds a training-only auxiliary task with 17 tissue fields to RK4.

Shared members reduce the total to **26 checkpoints: 14 refits and 12 originals**. Its **0.968 public / 0.958 private** was below G36-Gold privately. Both composition and fusion differ, so this is not an isolated comparison of soft mixing and hard routing.

6. Inference and deployment

Two T4 workers share DICOM decoding and reusable preprocessing, preserving checkpoint-specific inputs. FP16 and chunked encoding control memory. We verified member identities, local/Kaggle visible-test predictions and class-wise fusion. Both offline packages completed scoring within the limit; runtime forecasts are not hidden-test measurements.

7. Experiments and results

Private scores are retrospective. Gold58 results below use non-Gold-trained models on a repeatedly used development set, not a blind test. Leaderboard precision is three decimals.

7.1 OAI targets and training recipes

All runs use BiT-50 depth, seed 2026 and fixed epoch-25 EMA. N/R identify RSNA targets; expert/teacher ratios apply only to mapped OAI cells.

Run	RSNA targets	OAI / epoch	Expert : teacher	Gold58	Public	Private
E1 baseline	N	None	—	0.9400	0.962	0.951
J-v3	N	2,000	70 : 30	0.9446	0.963	0.953
J-v3-all	N	9,440	70 : 30	0.9408	0.962	0.951
J-kd50-all	N	9,440	50 : 50	0.9453	0.963	0.952
RK4	R	4,000	50 : 50	0.9505	0.963	0.953
N3RK4-o18	R	4,000 through epoch 18	50 : 50	0.9523	0.963	0.953

Target mixing is the cleanest pair. J-v3-all and J-kd50-all differ in OAI target/weight paths: 70:30 versus 50:50, with label-dependent offsets recomputed. Local AUC rose from 0.9408 to 0.9453; both displayed leaderboard scores rose 0.001.

More OAI was not uniformly better. At 70:30, the full pool underperformed the 2,000 draw, although more data also meant more optimizer updates. RK4 changes targets and OAI budget. N3RK4-o18 adds auxiliary supervision and an OAI-free tail; its higher local AUC did not improve displayed leaderboard scores. These are not single-factor ablations of every component.

7.2 Late ensembles and Gold58 refits

Submission	Checkpoints	Gold58-refit checkpoints	Public	Private
A21, non-Gold	21	0	0.968	0.959
A28, seed-expanded	28	0	0.968	0.958
A35, seed-expanded	35	0	0.967	0.958
F44, mixed groups	44	0	0.968	0.959
G36, non-Gold	36	0	0.968	0.959
G36-Gold — selected	**36**	**27**	**0.968**	**0.959**
A21G, equal-weight refits	30	24	0.968	0.958
A21GT — selected	**26**	**14**	**0.968**	**0.958**

A28/A35 changed seed counts and flat-mean recipe shares together, so their lower private scores do not isolate a seed effect. G36 instead used fixed group budgets.

G36 and G36-Gold both display **0.959 private**, as do A21 and F44. This precision establishes neither a Gold-refit benefit nor a grouping advantage. G36-Gold was best among our two selected entries, delivering third place.

Code and reproducibility

Inference notebooks: G36-Gold and A21GT. Training code, annotation prompts, configurations, and model weights will be released in a follow-up update.

References

- Competition: RSNA Knee Abnormality Detection.
- External data: Osteoarthritis Initiative.
- Backbone references: Big Transfer (BiT), ConvNeXt, DINOv3, and CoAtNet.
- Model library: PyTorch Image Models (timm).

Appendix: Matrix20: exact configuration list

Five configurations per backbone family. This appendix describes the frozen model slots, not their Gold/non-Gold provenance. N and R are the label families described in the article. OAI=0 means no joint OAI training; all means the available pool.

Backbone	Resolution	Targets	OAI / epoch	Preprocessing
BiT	224	N	4000	native

Backbone	Resolution	Targets	OAI / epoch	Preprocessing
BiT	384	R	0	legacy
BiT	384	N	4000	native
BiT	512	R	4000	native
BiT	512	N	2000	native
ConvNeXt-S	224	R	2000	native
ConvNeXt-S	384	N	0	legacy
ConvNeXt-S	384	R	4000	native
ConvNeXt-S	512	N	all	native
ConvNeXt-S	384	N	2000	native
DINOv3	224	N	all	native
DINOv3	384	R	2000	legacy
DINOv3	384	N	4000	native
DINOv3	512	R	0	native
DINOv3	384	R	all	legacy
CoAtNet	224	R	0	native
CoAtNet	384	N	all	legacy
CoAtNet	384	R	2000	native
CoAtNet	512	N	4000	native
CoAtNet	384	N	0	native